# Response to the recommender and reviewers

**Manuscript:** *Cross-Cultural Validation and Ideological Fairness of a Historical Perspective Taking Instrument: Evidence from Czech Secondary Students*

Dear Recommender and Reviewers,

Thank you for the careful and constructive reviews. I revised the manuscript to sharpen its measurement focus, add the requested reporting, and bound its claims to the single scenario, item format, sample, and ideological range studied. While revising, I also rechecked the analyses and corrected several reported results; these corrections are listed under META-01 at the end of this letter.

The responses below use the identifiers in the revision ledger. Section, table, and figure references refer to the revised manuscript and supplement; page and line numbers refer to the clean revised PDF. A tracked-changes PDF against the submitted version accompanies this letter.

**Location map for the revised files**

| Location | Contents |
|---|---|
| Abstract | Principal results and scope boundary |
| Sections 1–2 | Background, prior evidence, and research questions RQ1 and RQ2 |
| Sections 3–4 | Method, psychometric evidence, and focal relationships |
| Section 5 | Interpretation, limitations, and recommendations |
| Tables 1–5, Figure 1 | Factor-model fit, descriptives, DIF, invariance, correlations; observed relationships |
| Tables S1–S3, S3b | Item parameters, loadings, reliability, pairwise sample sizes |
| Tables S4, S4b | Instrument source populations; analysis-specific sample sizes |
| Table S5 | Preregistration mapping and deviations |
| Tables S6–S7 | NS-only invariance fit; HPT composite by ideology tertile |

Tables 2 and 3 of the submitted version (a loading summary and a reliability summary) were removed because they repeated the text and Tables S2–S3; the remaining tables were renumbered.

## Recommender

### REC-01: Single-scenario content-validity limitation

**Response:** I agree that this is the main limitation and have made it the first limitation in Section 5.6 (pp. 36–37, lines 702–710), under the heading *Content validity*, separate from sampling. It states that nine ratings of one Weimar voting dilemma are repeated judgments about the same content rather than nine independent samples of the domain, so the scores risk construct underrepresentation (Messick, 1995). It explains that scenario-specific knowledge, familiarity, and moral salience cannot

be separated from perspective taking, that the evidence supports interpretations of performance on this task rather than HPT as a general competence, and that the fairness findings are bounded in the same way. It calls for parallel item sets applied to several scenarios that vary in period, place, political valence, and familiarity. The Conclusion (Section 5.8, p. 39, lines 754–756) now opens with "For this single Weimar scenario".

**REC-02: Discriminant-validity claims**

**Response:** I replaced the former below-.85 rule with bounded internal-structure language. Section 4.1 (pp. 22–23, lines 458–463) reports the factor correlations as evidence that the item sets are distinguishable in this sample and states that this does not establish discriminant validity. Section 5.7 now lists, as recommendation (f) (p. 39, lines 750–752), testing whether the subscales differentially predict external criteria; protocol step 5 in Section 5.5 (p. 36, lines 698–699) makes the same point.

**REC-03: Construct names in results**

**Response:** Table 5 (formerly Table 7) and the Results narrative now use construct names, including the column headers "HPT composite" and "Contextualization". The primary six-item composite is defined once in Section 3.3.1 (p. 16, lines 326–330) and called "HPT composite" thereafter. Scale abbreviations are retained only where they identify named instruments.

**REC-04: Confidence intervals**

**Response:** Table 5 now reports Fisher-transformed 95% confidence intervals for every correlation. It states that the intervals are descriptive and do not adjust for clustering (p. 30, lines 582–587). Exact pairwise sample sizes are reported in Table S3b, and the values reproduce from `revision_reporting_analyses.R`.

**REC-05: Brief-scale consistency**

**Response:** I adopted your suggestion. Section 5.3 (pp. 33–34, lines 638–654) now opens with the breadth-precision trade-off and Loevinger's attenuation paradox: raising reliability does not necessarily raise validity, and three items that sample different considerations about one decision can serve the construct better than near-duplicates. One sentence acknowledges that the resulting precision is modest and limits individual-level interpretation. The section then states the consequence you asked for: modest reliability attenuates validity coefficients, so the knowledge correlation is a lower bound, and the ideology associations are attenuated in the same way, which limits the strength of the null fairness evidence. Table S3 reports raw alpha with standard-error intervals, polychoric alpha, and omega for all scored constructs; the one-factor omega estimates for multidimensional composites are identified as descriptive.

## Reviewer 1

**R1-01: Abstract findings**

**Response:** The abstract (pp. 1–2, lines 10–22) now gives the primary knowledge and ideology correlations with 95% confidence intervals and states that no DIF was detected with this sample and procedure. It reports that multi-group CFA fit-index changes were compatible with scalar invariance and qualifies this evidence because the extreme groups are small and clustering was not mod-

elled. Sections 4.9, 5.1, and 5.8 (pp. 29–30, lines 572–578; pp. 32–33, lines 622–625; p. 39, lines 760–762) report the authoritarianism–reversed-presentism association as an unadjusted, hypothesis-generating result that was not subjected to the preregistered multiplicity procedure.

**R1-02: Organizing contribution and pathways**

**Response:** I agree that the protocol is the more useful contribution. The adaptation evidence and staged fairness evaluation now organize the paper from the research questions through the Conclusion. The former discussion of pathways (three paragraphs in Section 5.5 and one in Section 5.4) is condensed into a single paragraph in Section 5.4 (p. 35, lines 669–679). It lists the explanations the data cannot distinguish (a genuine null, cancelling pathways, a format effect, and task-demand compliance) and points to the response-process evidence in limitation 8 that could separate them. I also removed the phrase "the instrument's resistance to attitudinal contamination", which overstated the evidence.

**R1-03: Ideology mechanism and construct boundary**

**Response:** Sections 2.1 and 2.3 now distinguish contextual reconstruction from attitude-congruent endorsement, reduced integrative complexity, task-demand shortcuts, moral evaluation, and affective engagement. A construct-boundary paragraph (p. 11, lines 220–227) explains when affect and moral engagement may be target-relevant under broad historical-empathy accounts and when option matching or attitude congruence is construct-irrelevant under the narrower cognitive score interpretation used here. Section 2.3 (p. 10, lines 198–204) now also explains concretely why the ideology question arises for this instrument: every contextualization option describes Hannes as likely to vote for the NSDAP and every present-oriented option describes him as rejecting it, so the POP–CONT continuum coincides with the direction of the political choice (Wilschut & Schiphorst, 2019).

**R1-04: Hartmann and Hasselhorn source**

**Response:** The manuscript now cites Hartmann and Hasselhorn (2008) throughout. The Czech version was translated from that published English article, as the submitted version stated; Section 3.1 (p. 13, lines 268–272) says so and mentions Hartmann's (2008) dissertation once, as the instrument's original development source. Section 3.3.1 (pp. 15–16, lines 316–319) and the adaptation record in the reproducibility package were made consistent with this description.

**R1-05: Author-date citations**

**Response:** The Hartmann/Hasselhorn and Lee/Ashby citation keys were repaired. The rebuilt manuscript renders the authors' names and years rather than publication titles.

**R1-06: Validation evidence map**

**Response:** Section 2.1 (pp. 6–7, lines 124–140) now distinguishes the original German PCA result from the separate latent-class analysis and from the correlated three-factor CFA tested here. It reports successful Dutch reuse of the Nazi scenario while noting weaker reliability and validity evidence for transfer to a slavery scenario and later evidence that students sometimes misunderstood response options. Section 2.1 also cites Wilschut and Schiphorst (2019) as a second transfer study: with two new scenarios and 69 Dutch Grade 10 students, principal components did not recover

the three response categories and total-score reliability was low. Claims that the instrument had been extensively validated or that the Czech CFA replicated an original three-factor structure were removed.

**Reliability remark ("the issues with reliability are still present"):** As you note, reliability problems recur. Huijgen et al. (2014) and Wilschut and Schiphorst (2019) report weaker reliability when the format is moved to new content, and Section 5.3 (p. 34, lines 645–649) now places the Czech estimates against these: the Czech composite estimates (omega = .66–.70) are modest but higher than the total-score alphas of .22–.50 in the new-content adaptations, consistent with a structure that depends on the original scenario.

**School grades:** In this sample, history grade correlated with the HPT composite ($r$ = -.15, $p$ = .010; the negative sign reflects the Czech scale, on which 1 is the best grade). Section 4.9 (p. 29, lines 567–570) now notes that this resembles the original German result rather than the Dutch applications, in which HPT scores were not significantly correlated with history grades.

**R1-07: Historical action in the scenario**

**Response:** I compared the German source, published English article, and administered Czech instrument. Sections 1 and 3.3.1 (pp. 3–4, lines 61–64; pp. 15–16, lines 316–319) now state that Hannes discusses Germany's crisis and the 1930 election and that students judge explanations including whether he could vote for an anti-democratic party such as the NSDAP. The manuscript no longer describes joining an extremist movement, and the illustrative statement in Section 2.3, which was not an instrument item, was replaced by a description of the actual option alignment (p. 10, lines 198–204; see R1-03).

**R1-08: Czech history-education context**

**Response:** Section 2.2 (pp. 8–9, lines 169–176) now describes the lower-secondary framework's coverage of the 1920s and 1930s and its source-use expectation, with page-specific citation to the 2023 framework, and states that HPT is not named as a separate competence and that curriculum documents establish intended rather than enacted provision. Because no equivalent upper-secondary curriculum-document audit was conducted, the documentary claim is limited to the lower-secondary subsample and distinguished from teachers' eligibility confirmation at both levels.

**R1-09: Other-scale consistency and suitability**

**Response:** Section 4.3 (p. 23, lines 472–477) and Table S3 report reliability for FR-LF, KSA-3, historical knowledge, SDR-5, and the combined ideology score. Table S4 audits the original source populations. The manuscript now states (p. 17, lines 347–352) that the German FR-LF surveys included respondents aged 14 and above but were predominantly adult, while KSA-3 and SDR-5 were developed with adults; none directly validates a Czech adolescent version.

**R1-10: Results emphasis**

**Response:** A short roadmap at the opening of Results (p. 21, lines 425–429) now maps Sections 4.1–4.5 to RQ1, Sections 4.6–4.8 to RQ2, and Sections 4.9–4.10 to the focal relationships. Subsection

openings state the principal result before technical detail. Detailed item loadings, reliability estimates, source populations, and analysis-specific sample sizes are in the supplement; Table 5 and Figure 1 foreground the principal relationships.

**R1-11: Prior item-rating and think-aloud work**

**Response:** Section 2.1 (p. 7, lines 137–139) now separates Huijgen and colleagues' transfer of the item-rating approach across topics from their later combination of closed ratings with think-alouds. I do not claim that the cited studies directly compared open and closed response formats. Limitation 8 in Section 5.6 (p. 38, lines 734–743) proposes that direct comparison, adding that written explanations have detected change that closed ratings in this format missed (Wilschut & Schiphorst, 2019).

**R1-12: Gender-moderation claim**

**Response:** I found no direct source supporting the earlier adolescent gender-moderation statement and removed it. Section 5.6 (p. 37, lines 720–722) now states only that the sample imbalance limits assessment of gender heterogeneity and that no gender interaction or gender DIF analysis was conducted.

**R1-13: Qualitative response-process design**

**Response:** Following your suggestion, limitation 8 in Section 5.6 (p. 38, lines 734–743) now specifies observable think-aloud indicators: use of contextual evidence and integration of conflicting considerations, moral evaluation and affective engagement, paraphrase or option matching, and explicit attitude-congruent endorsement. These indicators would help distinguish intended reconstruction from the competing pathways that Section 5.4 lists.

**R1-14: Comparative studies**

**Response:** Section 2.1 (p. 7, lines 131–145) names and describes the verified Huijgen studies and Wilschut and Schiphorst (2019) rather than referring generically to comparative studies. Claims unsupported by direct use of the instrument were removed. I also cite a recent Turkish qualitative case study after full-text verification: it involved sixth- and eighth-grade students and used the Hartmann and Hasselhorn components as deductive analytic categories. I do not treat it as evidence of direct nine-item administration or psychometric performance.

## Reviewer 2

**R2-01: Research questions and predictions**

**Response:** Section 2.5 (pp. 12–13, lines 253–264) now presents two explicit research questions, labelled RQ1 (psychometric evidence for the Czech adaptation) and RQ2 (whether scores are affected by political attitudes). The labels are reused in the Results roadmap and the Summary of Findings (Section 5.1). A concise disclosure identifies post-registration additions and omitted registered tests; Table S5 provides the full mapping of the six registered hypotheses.

**R2-02: Participant and school descriptives**

**Response:** Section 3.2 (p. 15, lines 309–313) now gives participant-level ideology and knowledge medians alongside school distribution, gender, and grade level; Section 4.4 (p. 24, lines 484–490) provides the fuller distributions and floor and ceiling percentages. No defensible socioeconomic measure was collected. Sections 3.2 and 5.6 (p. 37, lines 723–726) state that socioeconomic coverage and associated selection bias cannot be characterized rather than inferring status from school location. Table S4b reports analysis-specific sample sizes.

**R2-03: Baseline presentation of key relationships**

**Response:** Table 5 reports unadjusted correlations with confidence intervals, and Figure 1 displays the continuous ideology-HPT and knowledge-HPT relationships with sample sizes and uncertainty. I also added Table S7, which reports the mean HPT composite with a 95% confidence interval for the low, middle, and high tertiles of the composite ideology score; a sentence in Section 4.9 (p. 30, lines 588–590) points to it. The continuous display remains primary because tertiles are arbitrary cut-points of a continuous score. The Results distinguish these descriptive patterns from the adjusted multilevel models.

## Additional corrections

**META-01: Changes to reported results since the submitted version**

While revising, I rechecked the analyses against the archived data and code. The table lists every change to a reported result or disclosure relative to the submitted version.

| Item | Submitted version | Revised version |
|---|---|---|
| Three-factor CFA fit | CFI .999, RMSEA .011 (unscaled) | CFI .975, RMSEA .039 (scaled WLSMV); described as good, not near-perfect |
| Bifactor vs. three-factor | $\Delta$CFI .001, "no meaningful improvement" | $\Delta$CFI +.018; fit favors the bifactor model; structural claims softened |
| MG-CFA admissibility | Heywood cases in the Low group | All solutions admissible; Heywood claim withdrawn |
| Focal FR-LF coefficient | $\beta = .012$, $p = .863$ | $\beta = .016$, 95% CI [-.117, .149], $p = .817$, $n = 282$ (one scoring rule applied throughout); singular fit disclosed |
| Design-effect argument | DEFF 1.56/2.16 used to justify inference | Removed (it combined classroom size with school ICCs) |
| ICCs | School level only | School / class-within-school / total |
| DIF description | Uniform and non-uniform tests | Omnibus joint test; raw POP in the GRM disclosed |
| Modification index | Read as loading non-invariance | Read as a POP2 cross-loading diagnostic only |

| Item | Submitted version | Revised version |
|---|---|---|
| Authoritarianism-presentism | Not reported | Reported as exploratory and unadjusted |
| Preregistration disclosure | TOP form: one deviation (TOST bound) | Table S5 lists omitted registered analyses, post-registration additions, and the extension of the registered tertile-group MG-CFA from the CONT items to all nine items |
| Repository | OSF project | Zenodo, https://doi.org/10.5281/zenodo.23494026 |

None of these changes alters the substantive conclusion: historical knowledge remains the only consistent predictor, and no ideology effect or DIF was detected. Two corrections deserve particular mention. First, the submitted TOP form listed a single deviation from the preregistration; that statement was incomplete. Table S5 now discloses every omitted registered analysis (including the MIMIC model, the knowledge moderation tests, the full covariate set, one-tailed tests, and the Benjamini-Hochberg correction) and every post-registration addition, and the revised TOP form refers to it. Second, because OSF has discontinued project storage, the data, code, and documentation now reside in a Zenodo record; the preregistration remains at https://osf.io/zsngy.

Sincerely,

Juda Kaleta

Charles University, Faculty of Arts, Institute of History, Czech Republic

juda.kaleta@ff.cuni.cz